

Pavani Goli

Data Engineer

314-254-3918 | pavani.g@mymailkeeper.com | Saint Louis, MO

Professional Summary

- Around 5 years of experience as a Data Engineer, specializing in optimizing scalable data pipelines for efficient data processing and analytics with cloud and on-premise data solutions to drive data-driven insights and operational efficiency.
- Experience in Apache Spark, Apache Kafka, Apache Flink, Hadoop, Hive, HDFS, and other big data technologies for scalable data processing, real-time streaming, and efficient data storage solutions.
- Expertise in Azure technologies, including Azure Datafactory, Data Lake Storage, Databricks, and Azure Synapse, for scalable data processing and analytics, also in data visualization tools like Tableau and Power BI to transform complex data into actionable insights.
- Skilled in Matplotlib for data visualization, Snowflake for cloud data warehousing, and version control using Git and GitLab, as well as database management with SQL Server and PostgreSQL.

Skills

Programming Language: Python, R, Scala, SQL, NoSQL

IDE's: Visual Studio Code, PyCharm, Jupyter Notebook

Big Data Ecosystem: Apache Spark, Apache Kafka, Apache Flink, Hadoop, Hive, HDFS, MapReduce

Cloud: Azure (Data Lake Storage, Data Factory, Azure SQL databases, Databricks, Azure Synapse, Blob Storage)

Visualizations: Tableau, Power BI, Looker, Excel, Alteryx

Packages: NumPy, Pandas, Matplotlib, Seaborn, PySpark

ETL and Data Processing: SSIS, Informatica, DBT, Apache Airflow, Talend, Snowflake

Version Control & Database: Azure Devops, GitHub, Git, Gitlab, SQL Server, PostgreSQL, Cassandra, MySQL, HBase

Education

B.Tech in Computer Science & Engineer

Jawaharlal Nehru Technological University Hyderabad, India

Work Experience

CitiGroup, IL | Data Engineer

Aug 2024 – Present

- Implemented scalable Spark-based data pipelines to efficiently transform and process raw, semi-structured, and structured data, ensuring seamless integration into BI systems, which led to a 45% improvement in data processing speed.
- Designed scalable data pipelines in Azure, optimizing ETL processes and reducing data processing time by 40% using Azure Data Factory, Synapse Analytics, and Databricks. Ensured data integration across multiple sources, improving reporting efficiency.
- Implemented Power BI dashboards connected to Azure SQL and Data Lake, enabling real-time data visualization and improving executive decision-making speed by 25%. Optimized DAX queries and data models, reducing report load times by 35%.
- Applied modular and reusable DBT transformations by leveraging Jinja macros, refactoring SQL logic, and implementing best practices for data modeling, which resulted in an improvement in data pipeline efficiency.
- Enhanced ETL workflows by rigorously testing and debugging SQL scripts and Python code, optimizing data transformation processes and improving processing efficiency by 35%. Implementing error handling, data validation, and real-time monitoring led to more reliable data integration with downstream systems, ensuring smooth data flow and reducing failure rates.

Zensar Technologies, India | Data Engineer

Jun 2019 – Feb 2021

- Deployed Apache Flink-based stream processing applications to efficiently transform, aggregate, and enrich high-velocity data from multiple sources, ensuring seamless integration with BI systems.
- Integrated Azure Data Lake with multiple on-premises and cloud-based data sources, enabling seamless ingestion, and retrieval of structured and unstructured data partitioning, resulting in a 50% enhancement in data accessibility.
- Built Power BI datasets by implementing advanced data modeling measures using DAX and Power Query, ensuring efficient data transformation and aggregation up to 40% increase in report accuracy and flexibility.
- Constructed scalable data lake on Azure Data Lake Storage Gen2, handling over 50TB of structured and unstructured data, improving data accessibility by 40% and enhancing scalability for analytics and machine learning workloads, resulting in a 30% reduction in data retrieval time for key business insights.
- Executed exploratory data analysis on data stored in Snowflake using Python libraries like Pandas and Matplotlib to identify trends and patterns for further investigation.

TopHat Software Technologies, India | Jr. Data Engineer

Jan 2017 – May 2019

- Developed PySpark ETL pipelines on daily data, achieving a 30% reduction in processing time by implementing data partitioning, broadcasting variables, and optimizing Spark SQL queries for improved performance.
- Generated a scalable Data Lake Storage solution to efficiently process structured and unstructured data from multiple enterprise sources, improving data retrieval performance and reducing storage costs by 40%.
- Optimized Power BI datasets and reports, reducing memory consumption by 20% through effective use of aggregations, incremental refresh, and column store indexing in Azure SQL, resulting in faster query performance.
- Boosted Snowflake storage and compute cost management strategies by leveraging automatic scaling, resource monitoring, and query performance tuning and materialized views to efficiently allocate reducing 40% unnecessary compute consumption.
- Executed high-performance Hive queries to efficiently process and analyze structured and unstructured data, leveraging advanced techniques such as partitioning, bucketing, indexing, and vectorized query execution.